



Lab 02 – Android UI & Event Programming

Lab Objectives

By the end of this lab session, the student should be able to:

- Become familiar with the Android Studio environment.
- Create and manipulate basic graphical components in Android.
- Program in Java for Android.
- Handle events (click actions).

1 Exercise 1: Currency Converter

Objective

Create an application that converts:

$$1\text{£} = 280 \text{ DZD}$$

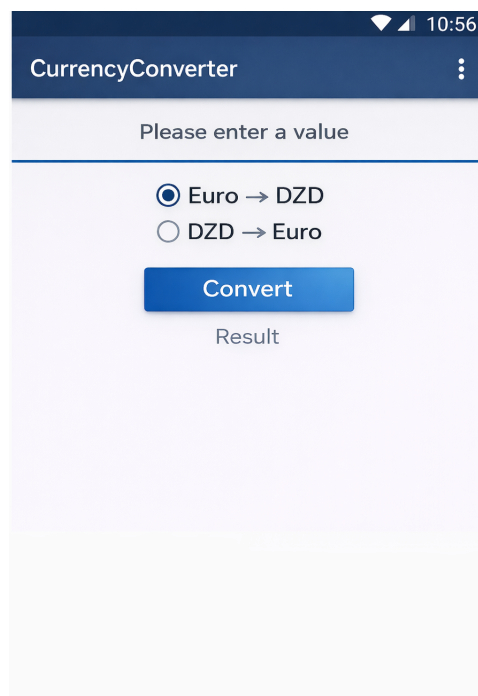
$$1 \text{ DZD} = 0.0034\text{£}$$

Steps to Follow

1. Creating the XML Interface

The interface must contain:

- One **EditText** to enter the value.
- Two **RadioButtons**:
 - Euro → DZD
 - DZD → Euro
- One **Button**: Convert
- One **TextView**: Result





2. Java Logic

Important steps:

1. Retrieve the entered value.
2. Check if it is numeric.
3. Verify which RadioButton is selected.
4. Apply the corresponding formula:

If Euro $\rightarrow$ DZD: $amount \times 144.92$

If DZD $\rightarrow$ Euro: $amount \times 0.0069$

2 Exercise 2: Conjugator (1st Group – Present Tense)

Objective

Conjugate a 1st group French verb (ending in **-er**) in the present tense.

Rule Reminder

Remove **-er** and add:

Pronoun	Ending
Je	e
Tu	es
Il/Elle	e
Nous	ons
Vous	ez
Ils/Elles	ent

Steps to Follow

1. Create an EditText to enter the verb.
2. Add a **Conjugate** button.
3. Verify that the verb ends with **-er**.
4. Remove the last two letters.
5. Add the corresponding endings.





Example : If the user enters: parler

Root: parl

Result:

- Je parle
- Tu parles
- Il parle
- Nous parlons
- Vous parlez
- Ils parlent

3 Exercise 3: Price Calculator

Formulas

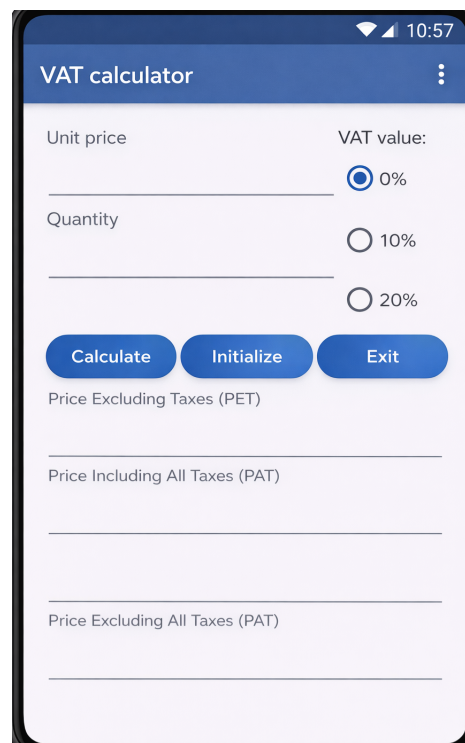
$$PET = UP \times Qt$$

$$PAT = PET + (VAT \times PET)$$

Steps to Follow

Constraints

- VAT values: {0%, 10%, 20%}
- Default value: 0%
- Use only **LinearLayout**
- Buttons:
 - Calculate
 - Initialize
 - Exit





Programming Steps

1. Retrieve UP and Qt.
2. Check that they are numeric.
3. Calculate PET.
4. Calculate PAT according to the selected VAT.
5. Display the results.

Buttons

Initialize:

- Clear all fields.
- Reset VAT to 0%.

Exit:

- Close the application using `finish()`.